Validation of a Mortgage Probability-of-Default Model

Champion (gradient-boosted trees) vs challenger (WoE scorecard), Freddie Mac single-family loan-level data

Qavi Inan

October 22, 2026

Executive summary

Model. A 36-month probability-of-default (PD) model for conforming, fixed-rate, full-documentation mortgages at origination, built on Freddie Mac's public single-family loan-level *sample* dataset (50,000 loans per vintage). Inputs are sixteen origination-time borrower and loan attributes (§4); geography, seller and vintage are excluded by design. A LightGBM champion with monotone constraints is compared with a weight-of-evidence logistic scorecard challenger and a FICO-only baseline.

Intended use. Rank-ordering of new originations for acquisition screening and risk-based review, and point-in-time PD estimation within the post-2009 underwriting regime. Not intended for capital, stress-testing or downturn pricing without a macro overlay.

Overall rating: Fit for intended use, with conditions.

Headline evidence.

- **Discrimination holds out of time.** Champion AUC 0.795 (95% CI 0.780–0.809) on 97,449 loans / 755 defaults from the 2017–2018 vintages, against 0.790 for the scorecard and 0.739 for FICO alone. The champion's edge over the scorecard is small but significant (ΔAUC +0.005, DeLong p = 0.008). The riskiest decile captures 41% of OOT defaults.
- **Calibration.** Pooled OOT observed/predicted = 0.96; by vintage 2017 1.31, 2018 0.72. On the 2006, 2007, 2008 crisis vintages the model still ranks (AUC 0.783) but predicts 3.2% against 6.2% observed: an under-prediction of 1.9×. The model has no macro inputs and has never seen a downturn.
- **Monotone constraints.** With LightGBM's `advanced` method, predictions move against the declared direction for 11.2% of loans on ltv, 4.5% of loans on dti, 1.5% of loans on fico, 0.1% of loans on orig_rate (LightGBM issue #5358). The champion uses `intermediate`.
- **Target definition.** Without the hardship guard, 36-month default rates for 2017, 2018, 2019 are 3.31%, 4.61%, 4.88% instead of 0.91%, 0.70%, 0.48%.

Findings. 1 High (0 open), 10 Medium, 2 Low (Section 11). The conditions on the rating are: no use for PD *levels* outside the 2009–2018 regime without a macro overlay; quarterly monitoring with the triggers in Section 12; replacement of the origination-rate input by a rate spread at the next redevelopment.

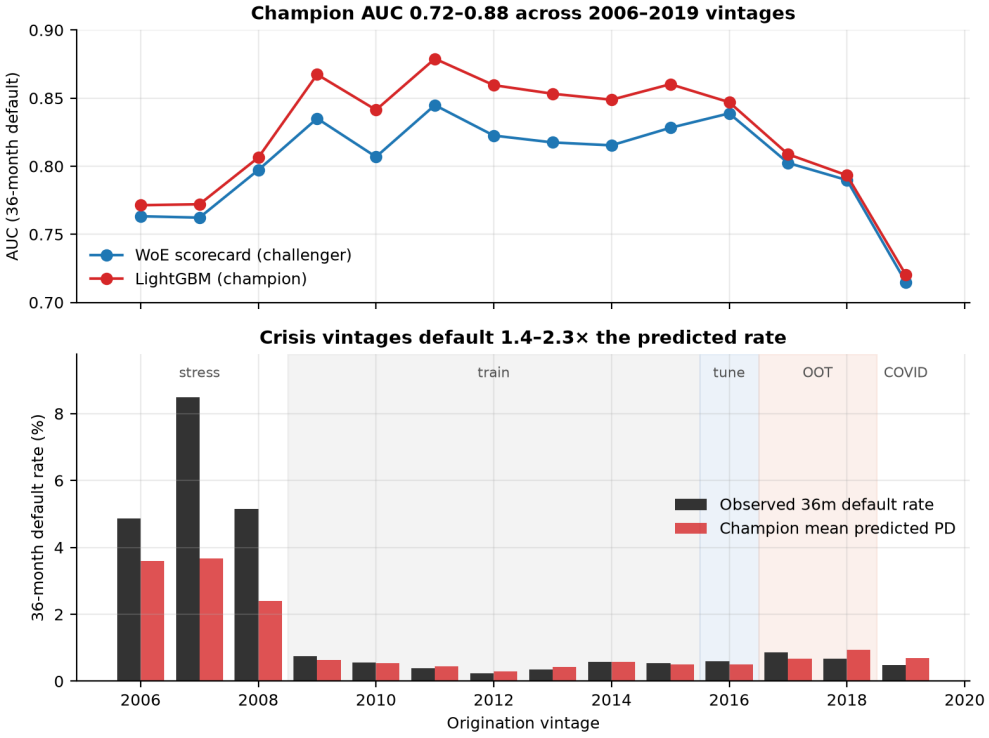


Figure 1: AUC and default rate by vintage.

1 Model overview, intended use and materiality

The model estimates the probability that a newly originated loan becomes 90 or more days delinquent, or terminates through a credit event, within 36 months of its first payment. It is an *acquisition-time* model: every input is known at origination and no performance information is used. Population: fixed-rate, fully amortising, full-documentation conforming loans sold to Freddie Mac, excluding HARP/Relief Refinance loans, government loans, ARMs and non-standard programs (these are absent from Freddie's Standard dataset or excluded here).

Under SR 26-2 the intensity of validation follows *materiality*, the product of model exposure and purpose. For the purposes of this exercise the model is treated as it would be if it screened a conforming purchase book: material by exposure, moderate by purpose (screening and review, not capital). That places it in the tier where independent validation with outcomes analysis, benchmarking and sensitivity testing is expected before use. The same person built and validated this model, so the review is not independent in SR 26-2's sense.

2 Regulatory context

This validation follows the *Supervisory Guidance on Model Risk Management* issued jointly by the Federal Reserve, OCC and FDIC on April 17, 2026 (Federal Reserve SR 26-2; OCC Bulletin 2026-13), which supersedes SR 11-7 (2011) and SR 21-8 and rescinds OCC 2011-12. SR 26-2 names three components of validation, **conceptual soundness**, **outcomes analysis** and **ongoing monitoring**, and frames the whole exercise through *effective challenge* by objective experts with the standing to effect change. Sections 5–8 of this report map onto those components; Appendix D gives the cross-walk to SR 26-2 paragraphs and to the SR 11-7 headings most readers know.

Footnote 3 of SR 26-2 carves out generative and agentic AI: "Generative AI and agentic AI models are novel and rapidly evolving. As such, they are not within the scope of this guidance. […] However, the principles described in this guidance apply to traditional statistical and quantitative models and non-generative, non-agentic AI models." A gradient-boosted PD model is squarely in scope.

Two further references shape the tests: the Basel Committee's *Studies on the Validation of Internal Rating Systems* (Working Paper 14, 2005) for the discrimination and calibration test catalogue, and the PRA's SS1/23 Principle 4 as an independent checklist of what a validation must cover. Fair-lending context comes from ECOA/Regulation B and CFPB Circular 2022-03.

3 Data

Source and lineage. Freddie Mac Single-Family Loan-Level Dataset, Sample files, Release 47 (July 2026), downloaded from Clarity Data Intelligence on 2026-09-12. Each vintage zip is hashed (SHA-256) in `data/manifest.json`; the file layout is verified programmatically against Freddie's July 2026 layout spreadsheet (31 origination fields, 35 performance fields) before any column is named. Fourteen vintages were ingested, 700,000 loans and 44,653,660 loan-months, with zero orphan records between the origination and performance files. Performance runs through 202603.

Target definition. `default_36m = 1` if within months 1–36 after first payment the loan is ever 90+ days delinquent (Freddie delinquency status ≥ 03, or RA), or terminates through a third-party sale, short sale/charge-off or REO disposition (zero-balance codes 02/03/09). Loan age is computed from First Payment Date, not Freddie's Loan Age field, which resets on modification. A 90+ month is **not** counted when the loan is in, or within 3 months after, a servicer-reported forbearance plan (Borrower Assistance Plan = F) or a disaster-related delinquency flag. Credit-event terminations always count. Loans are observable if they reach month 36 or pay off voluntarily before it; defects (ZB 96) and loans sold before month 36 without a default are excluded. The pure ever-90+ definition and the forbearance-only guard are retained as sensitivities.

Without the guard, the 2017, 2018, 2019 vintages show 36-month default rates of 3.31%, 4.61%, 4.88%, against 0.91%, 0.70%, 0.48% with it. Under the guarded definition the out-of-time vintages sit at 0.91% and 0.70%, close to 2016's 0.65%. This is a *data-relevance* issue in SR 26-2's sense and is carried as a Medium finding. The target definition and the train/tune split were revised after the out-of-time default rates were first examined.

Sample design. Vintage = sample file year (origination year; about 17% of each file's loans have a first payment in the following calendar year).

vintage	role	sample loans	observable	after HARP excl.	defaults	36m rate
2006	stress	50000	49773	49773	2417	4.86%
2007	stress	50000	49484	49484	4205	8.50%
2008	stress	50000	49620	49620	2553	5.15%
2009	train	50000	49849	44518	336	0.75%
2010	train	50000	49911	35315	197	0.56%
2011	train	50000	49895	34462	135	0.39%
2012	train	50000	49892	32526	74	0.23%
2013	train	50000	49920	35552	123	0.35%
2014	train	50000	49905	42623	243	0.57%
2015	train	50000	49965	45952	251	0.55%
2016	tune	50000	49934	47413	281	0.59%
2017	oot	50000	49879	48077	419	0.87%
2018	oot	50000	49883	49372	336	0.68%
2019	covid_exhibit	50000	49877	49847	239	0.48%

Train = 270,948 loans / 1,359 defaults; tune = 2016 (early stopping and hyper-parameter selection only); OOT = 97,449 / 755; stress = 148,877 / 9,175. The 2019 vintage's window spans the pandemic and is shown as a monitoring exhibit, not a test.

Data quality. FICO is missing on at most 0.1% of loans per vintage. DTI is missing on 2.1%–2.2% of 2006–2008 loans; from 2009 it is missing only on HARP loans. MSA is null on 9%–23% of loans (non-metro properties). HARP loans, 8%–35% of 2009–2015 originations, are excluded; this is the largest reduction in the waterfall and leaves 74–336 defaults per training vintage.

4 Model development summary

	Baseline	Challenger	Champion
Form	Logistic on Classic FICO (+ missing flag)	WoE scorecard: optbinning monotonic bins → L2 logistic	LightGBM, binary objective
Inputs	1	16	16
Constraints	—	Monotonic binning trend per input	`monotone_constraints` on FICO (−), LTV, CLTV, DTI, rate (+); method = intermediate
Selection	—	C = 1	Random search, 24 configs; early stop on tune AUC; best: 63 leaves, min_child_samples 500, lr 0.03, L2 0.0, 67 trees
Class handling	None (no resampling or re-weighting; see §5)	None	None

Inputs (information value from the scorecard binning, training sample): fico (0.94), orig_rate (0.42), num_borrowers (0.30), dti (0.27), ltv (0.17), cltv (0.16), orig_term (0.14), loan_purpose (0.07); the remaining inputs each have IV < 0.06.

The search history in `outputs/model_card.json` shows tune AUC in a narrow 0.840–0.847 band across all 24 configurations, i.e. the result is not sensitive to the hyper-parameter draw. Seeds are fixed; `make all` reproduces every number in this report from the raw zips.

5 Conceptual soundness

Variable rationale. Each input has an expected economic direction: FICO (borrower credit history, −), LTV/CLTV (equity cushion and incentive, +), DTI (payment capacity, +), origination rate (risk-based pricing, +), number of borrowers (income diversification, −), loan purpose (cash-out refinance carries more risk than purchase), occupancy (investor/second-home), MI %, property type, channel, first-time-buyer flag, super-conforming flag, units, term and UPB. Geography (state, MSA, ZIP3), seller and servicer are **excluded** from the model and used only for segmentation and the fairness proxy: geography is a fair-lending risk, and seller/servicer identity is not a borrower attribute.

Method choice. A boosted-tree champion is justified only if it beats a well-built scorecard on out-of-time data and stays monotone in the inputs an underwriter would expect; both are tested in §6 and §7. Monotone constraints are essential for a credit model that will be explained to borrowers (Reg B adverse-action reasons) and to examiners.

No class-imbalance correction. The default rate is ~1%. Resampling or re-weighting would inflate predicted probabilities and destroy calibration, which is the property a PD model exists for (van den Goorbergh et al., 2022). Both models are trained on the natural prior.

Finding on the rate input. Origination interest rate has IV 0.42 and ranks 4th by permutation importance. The median rate moves between 3.75% and 4.88% across training vintages, so the input partly encodes *calendar time* rather than the borrower's risk-based spread. Recommendation: replace with the spread to the month's PMMS 30-year rate at redevelopment.

Monotone constraints. The champion's hyperparameters were refit with each LightGBM constraint method, and each input was moved across its grid for 5,000 OOT loans. Share of loans whose prediction moves against the declared direction:

input	basic (%)	intermediate (%)	advanced (%)
dti	0.0	0.0	4.5
fico	0.0	0.0	1.5
ltv	0.0	0.0	11.2
orig_rate	0.0	0.0	0.1

6 Outcomes analysis

6.1 Discrimination

sample	n / defaults	baseline	challenger	champion
train 2009–15	270,948 / 1,359	0.758 (0.745–0.771)	0.832 (0.821–0.842)	0.864 (0.855–0.873)
tune 2016	47,413 / 281	0.800 (0.772–0.824)	0.839 (0.818–0.858)	0.847 (0.826–0.867)
OOT 2017–18	97,449 / 755	0.739 (0.720–0.755)	0.790 (0.775–0.804)	0.795 (0.780–0.809)
stress 2006–08	148,877 / 9,175	0.738 (0.733–0.743)	0.775 (0.770–0.779)	0.783 (0.778–0.787)
COVID 2019	49,847 / 239	0.689 (0.655–0.721)	0.715 (0.682–0.747)	0.720 (0.689–0.752)

Both multivariate models beat the FICO-only baseline out of time (by 0.051 and 0.057 AUC; DeLong p $\leq$ 8e-19). The champion's edge over the scorecard, +0.005 on OOT (p = 0.008), is small. Train-to-OOT decay for the champion is 0.068 (0.864 $\rightarrow$ 0.847 $\rightarrow$ 0.795); the train-to-tune step is in-sample optimism, the tune-to-OOT step is regime drift shared by all three models. The decay exceeds the 0.03 tolerance and is carried as a Medium finding.

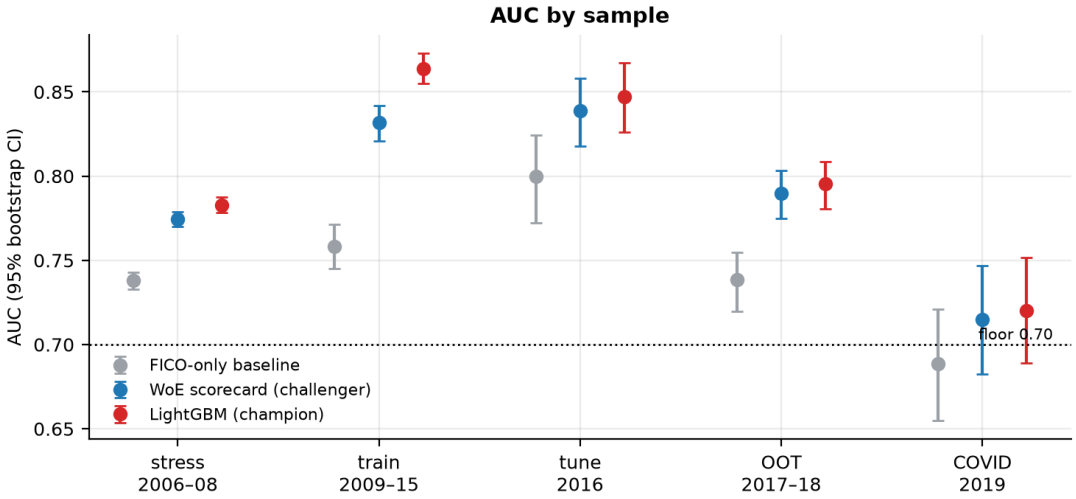


Figure 2: AUC by sample with bootstrap intervals.

Capture. The champion's riskiest decile holds 41% of OOT defaults at a 3.2% observed rate (lift 4.1×); the top two deciles hold 62%.

Segments (OOT, champion; segments with $\geq$ 100 defaults). The weakest within-segment ranking is fico_band = <660 (AUC 0.590, 95% CI 0.546–0.633), then fico_band = 660-699 (0.691). This is below the 0.65 segment floor and is carried as a finding. Full table in Appendix B.

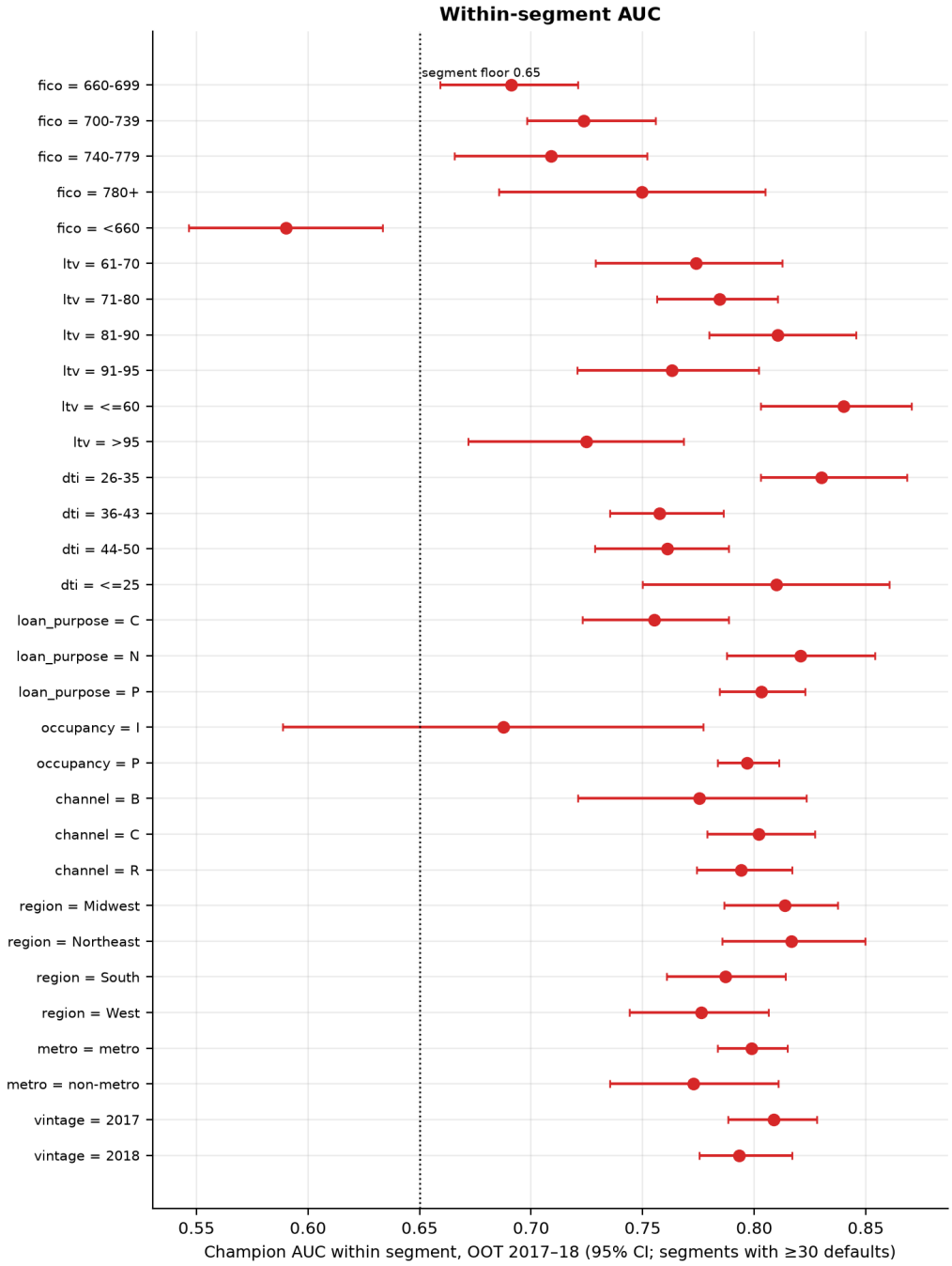


Figure 3: Within-segment AUC on the out-of-time sample.

6.2 Calibration

sample	observed	predicted	obs/pred	intercept	slope	slope SE	Brier skill	ECE
train 2009–15	0.50%	0.50%	1.002	1.617	1.353	0.025	0.021	0.002
tune 2016	0.59%	0.49%	1.201	2.075	1.400	0.065	0.013	0.003
OOT 2017–18	0.77%	0.80%	0.963	-0.180	0.970	0.035	0.010	0.001

sample	observed	predicted	obs/pred	intercept	slope	slope SE	Brier skill	ECE
stress 2006–08	6.16%	3.22%	1.916	0.200	0.815	0.009	0.044	0.029
COVID 2019	0.48%	0.70%	0.685	-1.558	0.742	0.060	-0.008	0.002

Pooled out of time, observed/predicted is 0.96 and the slope 0.97 ± 0.03; by vintage the ratio is 2017 1.31, 2018 0.72, so the pooled figure averages out errors in opposite directions. 3 of 10 OOT deciles have mean predicted PD outside the observed 95% Wilson interval. The slope is 1.35 on train and 1.40 on tune. Calibration-in-the-large is outside ±15% on: tune 2016, stress 2006–08, COVID 2019. The 2019 COVID exhibit (observed/predicted 0.69) is not a valid calibration test: forbearance, deferral and disaster relief suppressed defaults that the guarded target removes from the numerator.

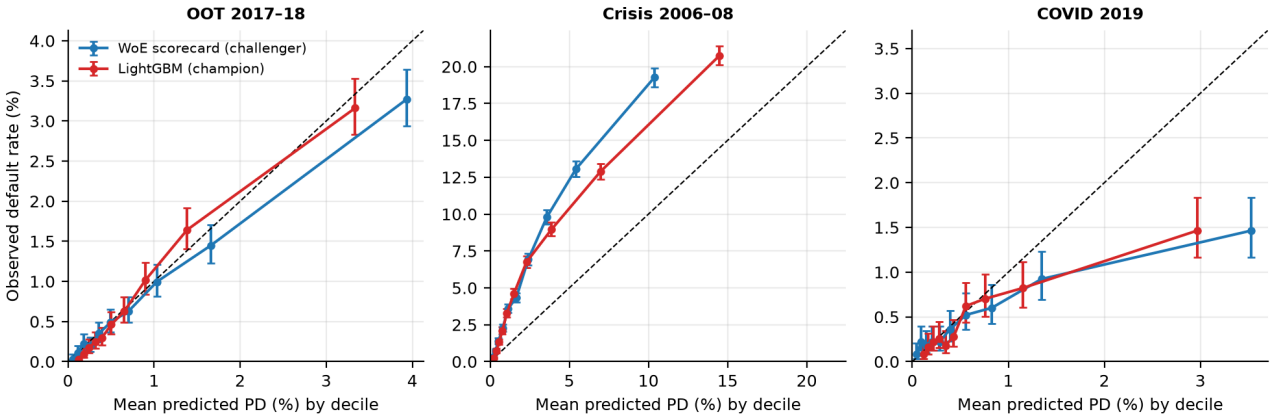


Figure 4: Reliability diagrams (deciles, Wilson intervals).

The model's inputs describe the borrower, not the economy; a 2006 borrower with a 720 FICO and 80 LTV looks like a 2014 borrower with the same attributes, and the model prices them the same. Rank-ordering survives (AUC 0.783) because relative risk is stable; the *level* does not. This is a design limitation, not a defect, but it bounds the intended use: PD levels from this model are point-in-time for the post-2009 regime and must carry a macro overlay for any capital, pricing or stress application.

6.3 Stability

vintage	role	n	score PSI	AUC	observed	predicted
2006	stress	49773	2.866	0.771	4.86%	3.59%
2007	stress	49484	2.801	0.772	8.50%	3.67%
2008	stress	49620	1.537	0.806	5.15%	2.39%
2009	train	44518	0.056	0.867	0.75%	0.63%
2010	train	35315	0.001	0.841	0.56%	0.54%
2011	train	34462	0.019	0.879	0.39%	0.45%
2012	train	32526	0.227	0.859	0.23%	0.30%
2013	train	35552	0.012	0.853	0.35%	0.42%
2014	train	42623	0.063	0.849	0.57%	0.58%
2015	train	45952	0.009	0.860	0.55%	0.51%
2016	tune	47413	0.006	0.847	0.59%	0.49%
2017	oot	48077	0.119	0.809	0.87%	0.67%
2018	oot	49372	0.385	0.793	0.68%	0.94%

vintage	role	n	score PSI	AUC	observed	predicted
2019	covid_exhibit	49847	0.109	0.720	0.48%	0.70%

Score PSI against the training distribution is green for training and tune vintages except 2012 amber (0.23); 0.12 for 2017 and 0.39 for 2018. The largest 2018 input shifts are orig_rate (CSI 0.87), loan_purpose (0.47) and ltv (0.19). Vintages over-predicted by more than 15%: 2012, 2013, 2018, 2019. The crisis vintages are far outside the development population (PSI above 0.25), which is the point of the stress test.

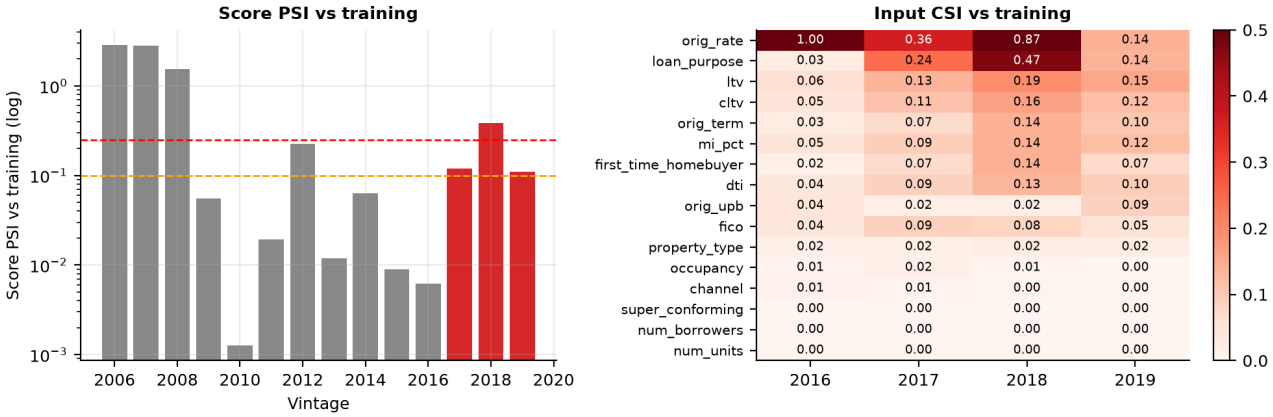


Figure 5: Score PSI by vintage and input CSI heat-map.

7 Sensitivity analysis

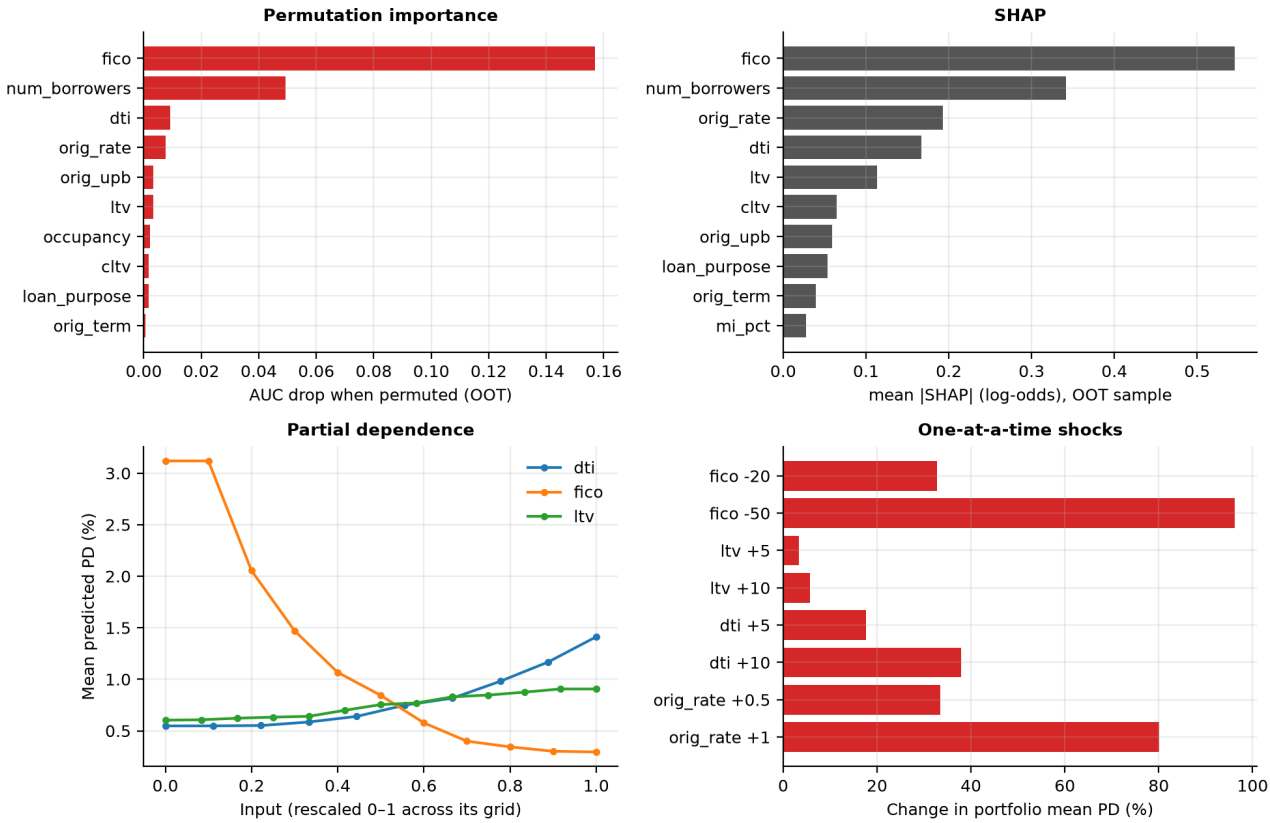


Figure 6: Permutation importance, SHAP, partial dependence and one-at-a-time shocks (champion, OOT).

FICO dominates (permutation AUC drop 0.157); num_borrowers (0.049), dti (0.009) and orig_rate (0.008) follow; SHAP gives the same top four. One-at-a-time shocks on the OOT book:

input	shock	portfolio mean PD	change
(none)	+0	0.80%	+0%
fico	-20	1.07%	+33%
fico	-50	1.58%	+96%
ltv	+5	0.83%	+3%
ltv	+10	0.85%	+6%
dti	+5	0.95%	+18%
dti	+10	1.11%	+38%
orig_rate	+0.5	1.07%	+33%
orig_rate	+1	1.45%	+80%

Champion monotonicity grid test on 5,000 OOT loans (FICO 620–820 by 20, LTV and DTI by 5, rate 2.5–7% by 0.5): maximum violation rate 0.0%.

8 Benchmarking and model recommendation

The champion's AUC lead over the scorecard ranges from +0.005 to +0.032 across samples. The scorecard is transparent (one coefficient per bin); the champion needs SHAP for reason codes. **Recommendation:** adopt

the champion for ranking, with the scorecard retained as the production challenger and the FICO-only model as the floor in monitoring. If the institution's adverse-action process cannot consume SHAP-based reasons, the scorecard's cost is 0.005 AUC, which is small enough that transparency should win.

9 Fairness / disparate-impact proxy analysis

Groups are tertiles of the loan's metro area non-white population share (ACS 2022 5-year, table B03002, CBSA or metropolitan division), with non-metro loans as a separate group; 90% of OOT loans matched an ACS geography. T1 (lowest share) is the reference. The illustrative policy cutoff declines the riskiest 10% of the OOT book (PD > 1.79%).

proxy group	n	defaults	observed	predicted	pred/obs	AUC	pass rate	AIR	SMD
T1	29490	196	0.66%	0.78%	1.176	0.808	90.7%	1.000	0.000
T2	29091	214	0.74%	0.79%	1.074	0.810	90.3%	0.996	0.007
T3	28928	241	0.83%	0.85%	1.016	0.776	88.9%	0.981	0.059
metro-unmatched	407	13	3.19%	0.88%	0.277	0.852	87.7%	0.967	0.092
non-metro/unknown	9533	91	0.95%	0.79%	0.830	0.773	90.3%	0.996	0.010

Proxy-feature audit. Association between each model input and the area demographic share (Spearman for numerics, correlation ratio for categoricals):

variable	type	assoc
orig_upb	numeric	0.256
super_conforming	categorical	0.162
property_type	categorical	0.149
dti	numeric	0.108
channel	categorical	0.102
ltv	numeric	-0.098

Reading. Lowest adverse impact ratio 0.97 (passes the 4/5ths screen). Standardised mean score difference, T3 vs T1: 0.06 SD. Within-group AUC 0.77–0.85. Predicted/observed by group: T1 1.18, T2 1.07, T3 1.02, metro-unmatched 0.28, non-metro/unknown 0.83. The input most associated with area demographics is orig_upb (0.26).

The adverse impact ratio is compared with the 4/5ths rule as a *screen*, and within-group calibration (pred/obs) is the fair-lending-relevant quantity for a pricing or limit model: a group that is systematically under- or over-predicted is being mis-priced. This is a geography-only proxy in the lineage of the CFPB's 2014 BISG methodology, without the surname component that carries most of BISG's power; it cannot support a compliance conclusion. A bank would run this with applicant-level protected-class data (HMDA for mortgages) and a documented proxy method for other products.

10 Limitations

- **Population.** Freddie Standard-eligible loans only: fixed-rate, full-doc, conforming, already approved. No rejected applicants exist in the data, so the model cannot be used for approve/decline without reject inference. HARP loans are excluded.
- **Sample.** 50,000 loans per vintage, not the full 1–2 million; segment-level estimates in OOT have wide intervals (Appendix B).

- **Target.** Ever-90+ in 36 months is a delinquency target, not a loss target; no LGD/EAD. The hardship guard depends on servicer-reported flags whose usage changed in 2020.
- **No macro inputs and no downturn in training** (2009–2015). PD levels are point-in-time for that regime.
- **Fairness** is a geography-only proxy: no applicant-level protected-class data, no surname component, no compliance conclusion.
- **Process.** Self-validation by the model's developer. The target definition and train/tune split were revised after the out-of-time default rates were first examined.
- **Freddie's dataset is "living"**: historical records can be restated between releases; the manifest pins Release 47.

11 Findings register and overall rating

Severity definitions: **High** – materially affects fitness for intended use; must be remediated or use restricted before reliance. **Medium** – limits use or reliability in identifiable conditions; remediate within one cycle, monitor meanwhile. **Low** – documentation, minor deviation, or observation.

ID	Severity / status	Finding	Evidence	Recommendation
F10	High / Remediated	Monotone constraints not enforced with LightGBM's 'advanced' method	Same hyperparameters: ltv 11.2%, dti 4.5%, fico 1.5%, orig_rate 0.1% of sampled loans move against the declared direction with 'advanced'; 'intermediate' gives 0%. See LightGBM issue #5358.	Champion uses 'intermediate'.
F01	Medium / Open	Champion train-to-OOT AUC decay exceeds tolerance	train 0.864 -> tune 0.847 -> OOT 0.795 (gap 0.068)	Part in-sample optimism (train-tune gap), part regime drift (tune-OOT). Tighten regularisation; monitor AUC quarterly; re-estimate on rolling window.
F02	Medium / Open	Weak rank-ordering in segment fico_band=<660	AUC 0.590 (n=4,500, defaults=150)	Investigate segment-specific drivers; consider segment overlay.
F03	Medium / Open	Calibration-in-the-large breach on stress (model under-predicts)	predicted 3.22% vs observed 6.16% (obs/pred = 1.92x)	Model has no macro inputs and never saw a downturn; PD level is point-in-time for the training regime only. Do not use for capital/pricing without a macro overlay or recalibration; add crisis-vintage stress to monitoring.
F04	Medium / Open	Calibration-in-the-large breach on covid_exhibit (model over-predicts)	predicted 0.70% vs observed 0.48% (obs/pred = 0.69x)	Observed defaults were suppressed by pandemic forbearance, deferral and disaster relief, which the hardship-guarded target removes from the numerator; 2019 is not a valid calibration test. Treat forbearance-heavy periods as a data-relevance trigger and suspend calibration monitoring on affected cohorts.

ID	Severity / status	Finding	Evidence	Recommendation
F05	Medium / Open	Calibration-in-the-large breach on vintage 2017 (model under-predicts)	predicted 0.67% vs observed 0.87% (obs/pred = 1.31x)	The pooled OOT ratio hides offsetting vintage errors; monitor by vintage.
F06	Medium / Open	Calibration-in-the-large breach on vintage 2018 (model over-predicts)	predicted 0.94% vs observed 0.68% (obs/pred = 0.72x)	The pooled OOT ratio hides offsetting vintage errors; monitor by vintage.
F07	Medium / Open	Calibration slope outside range	train 1.35, tune 1.40	Slope > 1: predictions are too compressed (early stopping at few trees). Recalibrate on the tune vintage.
F08	Medium / Open	Score PSI red on vintage 2018	PSI=0.385	Population shift vs development sample; confirm inputs (CSI) and recalibrate if persistent.
F11	Medium / Open	Original interest rate is confounded with calendar time	orig_rate: scorecard IV 0.42, #4 in OOT permutation importance; mean rate by training vintage ranges 3.60-4.96%.	Replace with the spread to the month's PMMS 30-year rate (FRED MORTGAGE30US) at redevelopment.
F12	Medium / Open	Target definition depends on servicer-reported hardship flags	36-month default rate with / without the forbearance and disaster guard: 2017 0.91% / 3.31%, 2018 0.70% / 4.61%, 2019 0.48% / 4.88%.	Keep both definitions in monitoring; treat forbearance-heavy periods as a data-relevance trigger.
F09	Low / Open	Crisis vintages are outside the development population	PSI 2006–08 = 2.87, 2.80, 1.54	Expected by design of the stress backtest; documents that the model would be extrapolating in a 2006-style book.
F13	Low / Open	Development sample excludes HARP loans and is the 50k-loan public sample	HARP is 8%-35% of 2009-2015 originations. Train has 1,359 defaults across 270,948 loans after exclusion.	State scope in intended use; use the Standard dataset for segment-level recalibration if the model is adopted.

Overall rating: Fit for intended use, with conditions. Conditions: (1) no reliance on PD levels outside the 2009–2018 regime without a macro overlay or recalibration; (2) quarterly monitoring per Section 12 with the hardship-guarded and pure targets both tracked; (3) replacement of the origination-rate input by a rate spread, and a fair-lending review with real protected-class data, before any policy use; (4) the fico_band = <660 segment to be reviewed with a segment overlay or a dedicated model.

12 Ongoing monitoring plan

Metric	Frequency	Green / Amber / Red	Action on Red
Score PSI vs development	Quarterly	< 0.1 / 0.1–0.25 / ≥ 0.25	Input CSI review; recalibrate if persistent 2 quarters
AUC on matured cohorts (12-month early-warning target as proxy)	Quarterly	≥ 0.70 and within 0.03 of development / – / below	Redevelopment trigger

Metric	Frequency	Green / Amber / Red	Action on Red
Calibration-in-the-large (obs/pred)	Quarterly	within ±15% / ±15%–25% / > 25%	Intercept recalibration; macro overlay review
Segment AUC (FICO, LTV, purpose, occupancy)	Semi-annual	≥ 0.65 where ≥ 100 defaults	Segment overlay
Share of 90+ events under forbearance / disaster flags	Quarterly	< 10% / 10–30% / > 30%	Data-relevance review; suspend calibration monitoring on affected cohorts
Monotonicity grid test	On every re-fit	0% violations	Block release
Full revalidation	Annual, or on any redevelopment	—	—

Appendix A: Data dictionary and cleaning log

Inputs (origination file, Release 47 positions): Classic FICO (1), First Payment Date (2), First-Time Homebuyer (3), MSA (5), MI % (6), Units (7), Occupancy (8), CLTV (9), DTI (10), UPB (11), LTV (12), Rate (13), Channel (14), Amortization Type (16), State (17), Property Type (18), ZIP3 (19), Loan ID (20), Purpose (21), Term (22), Borrowers (23), Seller (24), Super-Conforming (25), HARP (28). Sentinel codes (FICO 9999; LTV/CLTV/DTI/MI 999; units 99; categorical 9/99) are mapped to missing; categorical missing becomes an explicit "NA" level. Performance file fields used only for the target: Period, Delinquency Status, Zero Balance Code, Borrower Assistance Plan, Delinquency Due to Disaster. Unit tests cover every branch of the target definition (15 hand-built loan histories) and assert that no performance column other than the target enters the feature frame.

Appendix B: Segment tables (OOT, champion)

dimension	segment	n	defaults	obs_rate	mean_pred	AUC (95% CI)
fico_band	660-699	12923	232	1.80%	1.71%	0.691 (0.659–0.721)
fico_band	700-739	22722	197	0.87%	0.92%	0.724 (0.698–0.756)
fico_band	740-779	29511	112	0.38%	0.44%	0.709 (0.666–0.752)
fico_band	<660	4500	150	3.33%	3.30%	0.590 (0.546–0.633)
ltv_band	71-80	38951	236	0.61%	0.78%	0.784 (0.756–0.810)
ltv_band	81-90	11460	100	0.87%	0.93%	0.810 (0.780–0.846)
ltv_band	91-95	15007	179	1.19%	1.07%	0.763 (0.721–0.802)
dti_band	26-35	28123	160	0.57%	0.64%	0.830 (0.803–0.868)
dti_band	36-43	30535	283	0.93%	0.94%	0.757 (0.735–0.786)
dti_band	44-50	22614	248	1.10%	1.10%	0.761 (0.728–0.789)
loan_purpose	C	21790	192	0.88%	0.97%	0.755 (0.723–0.789)
loan_purpose	P	61791	471	0.76%	0.79%	0.803 (0.785–0.823)
occupancy	P	84926	703	0.83%	0.84%	0.797 (0.784–0.811)
channel	C	32077	275	0.86%	0.83%	0.802 (0.779–0.827)
channel	R	55635	414	0.74%	0.79%	0.794 (0.774–0.817)
property_type	PU	26855	168	0.63%	0.73%	0.802 (0.767–0.834)
property_type	SF	61964	512	0.83%	0.85%	0.794 (0.776–0.811)
first_time_homebuyer	N	73464	511	0.70%	0.75%	0.793 (0.777–0.809)
first_time_homebuyer	Y	23985	244	1.02%	0.99%	0.795 (0.766–0.819)
num_borrowers	1	51775	589	1.14%	1.06%	0.749 (0.731–0.768)
num_borrowers	2	45237	163	0.36%	0.52%	0.822 (0.792–0.847)
region	Midwest	22664	175	0.77%	0.81%	0.814 (0.786–0.837)
region	Northeast	12762	116	0.91%	0.77%	0.817 (0.786–0.850)
region	South	34200	288	0.84%	0.85%	0.787 (0.761–0.814)
region	West	27793	176	0.63%	0.77%	0.776 (0.744–0.806)
metro	metro	87916	664	0.76%	0.81%	0.799 (0.784–0.815)
vintage	2017	48077	419	0.87%	0.67%	0.809 (0.788–0.828)
vintage	2018	49372	336	0.68%	0.94%	0.793 (0.775–0.817)

Appendix C: Reproducibility

`make data && make target && make features && make train && make validate && make figures && make report` from the raw zips in `data/raw/`; seed 20260911; Python 3.12, LightGBM 4.7, optbinning, scikit-learn, DuckDB; dependency lock in `uv.lock`. Raw file SHA-256 digests in `data/manifest.json`. Champion hyper-parameter search history in `outputs/model_card.json`.

Appendix D: Guidance cross-walk

Report section	SR 26-2 (2026)	SR 11-7 (2011) lineage
1 Intended use, materiality	§II Purpose and scope; §III model risk, materiality	§III Overview
4 Model development	§IV Model development and use (testing, out-of-time)	§IV Model development, implementation and use
5 Conceptual soundness	§V Components of validation – conceptual soundness	§V Validation – evaluation of conceptual soundness
6 Outcomes analysis; 7 Sensitivity; 8 Benchmarking	§V – outcomes analysis; §IV testing (comparison of alternatives)	§V – outcomes analysis; ongoing monitoring (benchmarking, sensitivity)
12 Monitoring plan	§V – ongoing model monitoring	§V – ongoing monitoring
11 Findings register	§VI Governance – documentation, tracking of recommendations	§VI Governance, policies and controls
9 Fairness proxy	out of scope of SR 26-2; ECOA/Reg B, CFPB Circular 2022-03	—